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Court and controversy: patenting science in the nineteenth century†

PAUL LUCIER*

Whenever we attempt to draw a line dividing between the *sciences*, usually so called, and the *arts*, it results in distinctions, which are comparative, rather than absolute. In many branches of knowledge, the two are so blended together, that it is impossible to make their separation complete...Discovery is the process; invention is the work of art. So common, however, is the connexion of the two with each other, that we find both a science and an art involved in the same branch of study.

Jacob Bigelow
Elements of Technology (1829)

The history of artificial illumination cannot fail to impress upon us the difficulties in the way of the application of scientific discovery to the utilities of life.

Edward Frankland
‘On artificial illumination’ (1863)

In the autumn of 1851, on the occasion of the American Institute of New York’s annual fair, the Boston chemist and geologist Charles Jackson chose as the subject of his address the ‘Encouragement and Cultivation of the Sciences in the United States’. Playing on popular enthusiasm for science and technology, Jackson rehearsed the wondrous progress of the arts and the role of science in that progress. Science was the ‘Hand-maiden of the Arts’, and most assuredly the ‘maid of honor’, he declared, for science was the ‘progressive power’ which inspired new inventions. These were commonplace assumptions of the time, and surely no one in his audience would have disputed them.¹

It was in the second part of his address where Jackson diverged from the accepted text. His subject was scientific discovery, and his grievance concerned priority. ‘Men of science devote their lives to increasing the sum of human knowledge’, he reminded his listeners. They gave their discoveries freely to the world; recently, however, they had become victims of ‘crimes brought on by selfish ambition’. Jackson reported that valuable discoveries

† *Editor’s note.* This essay was one of the joint winners in the Society’s Singer Prize Competition, 1994.

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1 C. T. Jackson, ‘Encouragement and cultivation of the sciences in the United States, Twenty-Fourth Anniversary Address, before the American Institute, of the City of New-York, at the Tabernacle, on 16th of October, 1851’, *Transactions of the American Institute* (1851), 227–46.

made by men of science were stolen by 'mere speculator[s] in inventions'. These 'cunning and artful' impostors appropriated scientific discoveries by patenting them. This was a new and present danger to scientists and, more important, it threatened to sever the traditional link between science and the arts. In anger Jackson demanded, 'How can we protect the true discoverer?'²

Despite the heated rhetoric, historians of science and technology have understandably overlooked Jackson and for the most part the problems of priority and patenting. Instead scholars have discussed the interrelations of science and industry by focusing on the two periods of most rapid and profound social and economic change, usually organized under the headings of 'First Industrial Revolution' and 'Second Industrial Revolution'. Whether or not science played a substantive role in early industrialization is the subject of much historical debate;³ however, most scholars assert that within the following century the place of science in industry was secured. In fact, the 'Second Industrial Revolution' has been characterized in part by the direct application of science to industrial innovation.⁴ Comparatively little attention has been paid to the mid-nineteenth century – the period during which the industrial role of science moved from debatable to essential.⁵ Something akin to a natural convergence of interests among scientists and entrepreneurs seems to serve as the causal explanation for this dramatic change. Yet Jackson's ringing demand

2 Jackson, op. cit. (1), 237.

3 See, for example, H. E. LeGrand, 'Theory and application: the early chemical work of J. A. C. Chaptal', *BJHS* (1984), 17, 31–46; A. R. Hall, 'What did the Industrial Revolution in Britain owe to science?', in *Historical Perspectives: Studies in English Thought and Society in Honour of J. H. Plumb* (ed. N. McKendrick), London, 1974, 129–51; R. Porter, 'The Industrial Revolution and the rise of the science of geology', in *Changing Perspectives in the History of Science: Essays in Honour of Joseph Needham* (ed. M. Teich and R. Young), London, 1973, 320–43; C. C. Gillispie, 'The natural history of industry', *Isis* (1957), 48, 398–407; and J. Mokyr, 'Editor's Introduction: The new economic history and the Industrial Revolution', in *The British Industrial Revolution: An Economic Perspective* (ed. J. Mokyr), Boulder, CO, 1993, 78–83. Alternative points of view are put forward by B. W. Keyser, 'Between science and craft: the case of Berthollet and dyeing', *Annals of Science* (1990), 47, 213–60; C. Perrin, 'Of theory shifts and industrial innovations: the relations of J. A. C. Chaptal and A. L. Lavoisier', *Annals of Science* (1986), 43, 511–42; N. McKendrick, 'The role of science in the Industrial Revolution', in Teich and Young, as above, 274–319; M. C. Jacob, *The Cultural Meaning of the Scientific Revolution*, New York, 1988; and A. E. Musson and E. Robinson, *Science and Technology in the Industrial Revolution*, Manchester, 1969.

4 D. S. Landes, *The Unbound Prometheus: Technological Change and Industrial Development in Western Europe from 1759 to the Present*, Cambridge, 1969. Work on American science and industry includes: M. A. Dennis, 'Accounting for research: new histories of corporate laboratories and the social history of American science', *Social Studies of Science* (1987), 17, 479–518; J. K. Smith, Jr, 'The scientific tradition in American industrial research', *Technology and Culture* (1990), 31, 121–31; and L. S. Reich, *The Making of American Industrial Research: Science and Business at GE and Bell, 1876–1926*, Cambridge, 1985, 12–41. On the rise of science-based industry in Germany see, for example, Ernst Homburg, 'The emergence of research laboratories in the dyestuffs industry, 1870–1900', *BJHS* (1992), 25, 91–111; G. Meyer-Thurrow, 'The industrialization of invention: a case study from the German chemical industry', *Isis* (1982), 73, 363–81; J. J. Beer, *The Emergence of the German Dye Industry*, Urbana, 1959; and L. F. Haber, *The Chemical Industry during the Nineteenth Century: A Study of the Economic Aspects of Applied Chemistry in Europe and North America*, Oxford, 1958.

5 This point is also made by S. H. Mauskopf, 'Chemistry and cannons: J.-L. Proust and gunpowder analysis', *Technology and Culture* (1990), 31, 398–426. Recent work on mid-to-late nineteenth-century chemistry and the emerging chemical industry includes, A. S. Travis, 'Perkin's mauve: ancestor of the organic chemical industry', *Technology and Culture* (1990), 31, 51–82; and A. S. Travis, 'Science's powerful companion: A. W. Hofmann's

serves as a warning that the course of science in the transition from 'First' to 'Second Industrial Revolution' was anything but inevitable or smooth.

On the contrary, the relations of science and industry were highly contentious, and Jackson's address highlights the uncertainty and urgency felt by many scientists and inventors regarding the propriety and possibility of patenting scientific knowledge. To scientists it seemed that entrepreneurs were not simply applying, but rather appropriating, their discoveries. To inventors, the danger appeared to be an attempt on the part of scientists to interfere and perhaps to prevent them from laying claim to novelty and invention. For both scientists and inventors it was becoming increasingly obvious that the rhetoric of useful knowledge had come face-to-face with the reality that money (sometimes a great deal of money) might be made from such knowledge.

The difficulties in the relations of scientists and inventors became apparent in numerous patent cases of the mid-nineteenth century. In these contests the role that science could and should play in industrial innovation was addressed directly, and it was here that scientists and inventors wrestled over the issues of priority and property.⁶ In this paper, I examine a particularly contentious patent dispute over coal oil. As one of the most expensive and extensive controversies of the 1850s and 1860s, it involved British and American scientists, inventors, lawyers, judges and journalists. At issue was the question of who owned the process for making an illuminating oil from coal. At its most public level, it was a struggle over property rights, but at a deeper and more significant level, scientists and inventors wrestled over the control and use of knowledge. This involved the social demarcation between two types of knowledge producers (scientists and inventors) and the legal distinction between two types of knowledge (science and technology). The coal oil controversy reveals the relative success and failure in maintaining these classifications and in the process the contentious relations of science and industry.

THE COAL OIL CONTROVERSY

The controversy had its origins in a British patent granted to James Young (1811–83), chemical manufacturer and self-described practical chemist. Young had studied chemistry under Thomas Graham at Anderson's Institution in Glasgow in the 1830s and had acquired

investigation of aniline red and its derivatives', H. van den Belt, 'Why monopoly failed: the rise and fall of Société La Fuchsine', and W. J. Hornix, 'From process to plant: innovation in the early artificial dye industry', all in a special issue, 'Organic Chemistry and High Technology, 1850–1950', *BJHS* (1992), 25, 27–44, 45–64, 65–90. On the relations of geology and industry see P. Lucier, 'Commercial interests and scientific disinterestedness: consulting geologists in antebellum America', *Isis* (1995), 86, 245–67.

6 Several scholars have noted the importance of patents in this regard, but they have tended to concentrate on the late nineteenth and early twentieth centuries. See, for example, D. Noble, *America by Design: Science, Technology and The Rise of Corporate Capitalism*, Oxford, 1979, 84–109; and Reich, *op. cit.* (4). For a discussion of patents in eighteenth-century Britain, see C. MacLeod, *Inventing the Industrial Revolution: The English Patent System, 1660–1800*, Cambridge, 1988; and H. I. Dutton, *The Patent System and Inventive Activity during the Industrial Revolution, 1750–1852*, Manchester, 1984. For a survey of the literature on the relations of science and technology see J. M. Staudenmaier, *Technology's Storytellers: Reweaving the Human Fabric*, Cambridge, MA, 1985, 83–120.

some manufacturing experience in Lancashire in the production of bleaching agents for cotton textiles, where he had worked with James Muspratt and Charles Tennant (Tennant, Clow and Company).⁷ Young's interest in coal oils began with an invitation from his friend Lyon Playfair to examine a petroleum seepage in a Derbyshire coal mine. Playfair had experimented with the distillation of petroleum and was convinced that it could produce a marketable illuminant. Young was also interested in the commercial possibilities of petroleum, but apparently the yield of the seepage was too low and irregular to develop any long-term plans. Young later recalled how he reasoned that the petroleum was produced from coal which was heated and squeezed by the overlying strata. An 'artificial' petroleum, he thought, could be made in a similar fashion. He thus began his experiments on the distillation of coal and soon developed a reliable process for producing an illuminant using coal.⁸ In October 1850 Young patented both the process and the product of that process, an oil he called 'paraffine oil'.⁹ His patent reads:

Be it known that I, James Young, of Manchester, England, have invented improvements in the treatment of certain bituminous substances, and in obtaining products therefrom, and do hereby declare the following to be a full, clear and exact description of the same. My said invention consists in treating bituminous coals in such manner as to obtain therefrom an oil containing paraffine (which I call paraffine oil), and from which I obtain paraffine.¹⁰

Two years later, in March 1852, Young acquired an American patent for paraffine oils.¹¹

In the mid-1850s the American and British coal oil industries began to prosper as the demand for safe, clean illuminants grew. By that time Young had established three manufactories in Britain, making him the largest producer of coal oils.¹² In the United States, however, Young did not build a factory, nor did he export any paraffine oil. Instead American developments were being led by the North American Kerosene Company, which was founded by the Canadian chemist and geologist Abraham Gesner (1797–1864). Gesner

7 For more on Young see John Butt, 'James Young, Scottish Industrialist and Philanthropist' (Ph.D. dissertation, Glasgow University, 1964); and R. J. Forbes, *Studies in Early Petroleum History*, Leiden, 1958, 182–94. Young's papers are located in Strathclyde University Library, Glasgow; this essay, however, was completed before the author had a chance to consult them. On the interrelations of British academic chemistry and the chemical industry see R. Bud and G. K. Roberts, *Science versus Practice: Chemistry in Victorian Britain*, Manchester, 1984; and J. F. Donnelly, 'Representations of applied science: academics and chemical industry in late nineteenth-century England', *Social Studies of Science* (1986), 16, 195–234.

8 On the development of the British coal oil industry see J. Butt, 'Legends of the coal-oil industry (1847–1864)', *Explorations in Entrepreneurial History* (1964), 2, 16–31, and 'Technical change and the growth of the British shale-oil industry (1680–1870)', *Economic History Review* (1965), 17, 511–21. On Playfair's role see W. Reid (ed.), *Memoirs and Correspondence of Lyon Playfair*, London, 1899.

9 British Patent No. 13,292, 17 October 1850.

10 'The manufacture of coal oil. The first patent', *Scientific American* (12 February 1859), 14, 186; and 'Young versus Fernie', *The Law Times Reports* (1864), 10, 861–5.

11 US Patent No. 8833, 23 March 1852.

12 In partnership with Edward Meldrum and the respected metropolitan geologist Edward William Binney, Young established three paraffine oil manufactories: E. W. Binney & Company of Bathgate, Scotland; E. Meldrum & Company of Glasgow; and James Young & Company of Manchester. For more on Binney see J. Binney, *The Centenary of a Nineteenth-Century Geologist: Edward William Binney*, Taunton, 1912.

had patented a type of coal oil called Kerosene; and Kerosene was fast becoming the most popular illuminant in the United States.¹³

Young watched these developments with great anxiety, and in November 1856, he engaged the services of a New York City law firm in an attempt to secure settlements with American companies manufacturing coal oils using methods similar to his. Young treated the Americans heavy-handedly and demanded they pay a royalty on each gallon of their home-produced coal oil. American companies resisted what they considered to be an unlawful and greedy attempt by a foreigner to exact tribute from them, and for two years, at least, they successfully avoided or delayed Young's attorneys. Toward the end of 1858, however, as a boom in coal oil was getting underway, Young began to put more pressure on them. It was during a period of intense negotiation between Young's attorneys and American manufacturers in the spring of 1859 that *Scientific American* published a seemingly innocuous article on coal oil. In response to a 'great number of inquiries regarding the history of this invention', it reprinted the text of what it thought to be the first coal oil patent, that of James Young.¹⁴

The article caused 'considerable commotion' among American coal oil interests, who reacted with a defiant show of industrial patriotism. One Baltimore refiner asked if there were not 'better and much quicker processes of manufacturing coal oils and paraffine, all invented in this country, and which can be secured at less cost than those of English importation.'¹⁵ A patent lawyer from Washington, DC, suggested a method for 'breaking down' (or nullifying) Young's patent, based on the very fact that Young was British. Under US patent law, aliens were required to put on sale their discovery within eighteen months of issue of an American patent. Since Young had not established a manufactory or marketed a single gallon of his paraffine oil in the United States, his patent was void. *Scientific American* regarded this legal loophole as simply 'another illustration of the severity of our miserable system discriminating between foreign inventors and our own citizens'.¹⁶

What bothered Americans even more than Young's nationality were the broad claims to discovery made in his patent specification. Many were unclear as to what was covered by the patent and what was not. 'So far as we know', explained *Scientific American* in February 1859, '[Young] is the discoverer of the method of obtaining oil from coal and...this seems to cover the manufacture of all coal oils by distillation in essence and principle'.¹⁷ But an Ohio manufacturer thought it was 'sublime impudence for a man to claim the discovery of coal oil in essence and principle', and cited the chemical research

13 US Patent Nos. 11,203, 11,204, and 11,205, 27 June 1854. For more on Gesner and Kerosene see Kendall Beaton, 'Dr. Gesner's Kerosene: the start of American oil refining', *Business History Review* (1955), 29, 28–53; Butt, 'Legends' op. cit. (8); and Harold F. Williamson and Arnold R. Daum, *The American Petroleum Industry: The Age of Illumination, 1859–1899*, Evanston, 1959, 43–60.

14 'Young's coal oil patent', *Scientific American* (12 March 1859), 14, 221. The article generated a large response, and, under the rubric of 'The coal oil controversy', *Scientific American* published some fifteen articles on Young's patent, from March to December 1859.

15 'Young's coal oil patent', *Scientific American* (5 March 1859), 14, 213.

16 'Young's coal oil patent', op. cit. (14).

17 'The manufacture of coal oil. The first patent', op. cit. (10).

of Berzelius and Liebig as proof that the process 'had been thoroughly investigated and published to the world' before Young's patent.¹⁸ This issue would come to dominate the entire controversy: how does the law (British and American) distinguish between the essence and principle of an invention and a scientific discovery?

The legal terms 'essence' and 'principle' were ambiguous and ill-defined both in the United States and in Britain.¹⁹ In spite of this, or perhaps because of it, patent specifications were usually phrased in such a way as to encapsulate the broadest claims to invention, and case law in both countries recognized something called the principle of an invention.²⁰ This referred to a technological principle (such as how a machine worked), and not a scientific law, which was by definition not patentable.²¹ Thus, for instance, an inventor could not patent the law of gravity, but he could claim a right to the principle of falling bodies and by extension all machines that worked on that principle. Patenting a mechanical principle used in a machine for manufacturing cotton textiles, for example, was relatively simple because there was no naturally occurring process for cloth production. With regard to chemical processes, the distinction between scientific law and technological principle was not nearly so clear-cut. In Young's case, one of the questions was whether the process for producing coal oil was a chemical one, meaning scientific and natural, or a practical one, meaning technological and artificial. The legal waters were muddied still further by the imprecise criteria used to establish priority. Prior invention, such as the construction of a machine before the patent, was solid ground for nullifying the patent. But did scientific publication of a chemical process constitute grounds of prior invention? These were the issues at stake in the coal oil controversy.

THE AMERICAN CONTEXT

The coal oil controversy moved from the pages of *Scientific American* to the community of American scientists by way of a vicious exchange in print between two chemists: Frank Storer (1832–1914) and Thomas Antisell (1817–93). Antisell was an examiner at the Patent Office, in Washington, DC, where he had specialized in the growing field of chemical manufacturing. In 1859 he published a book entitled *The Manufacture of Photogenic or Hydro-Carbon Oils*.²² Besides being an explanation of coal oil production, it was an

18 'The coal oil controversy', *Scientific American* (26 March 1859), 14, 238.

19 For more on the US patent system see S. Lubar, 'The transformation of antebellum patent law', and K. J. Dood, 'Pursuing the essence of inventions: reissuing patents in the 19th century', both articles in a special issue 'Patents and Invention', *Technology and Culture* (1991), 32, 932–59, 999–1017. For the British patent system see Dutton, op. cit. (6), and MacLeod, op. cit. (6).

20 Controversies over the precision of patent specifications are as old as patents themselves. The earliest examination of this issue involved James Watt and his patented steam engine; see E. Robinson, 'James Watt and the Law of Patents', *Technology and Culture* (1972), 13, 115–39. For a discussion of the social construction of patents and patent specifications see C. Cooper, *Shaping Invention: Thomas Blanchard's Machinery and Patent Management in Nineteenth-Century America*, New York, 1991.

21 This point was made explicit by the Commissioner of Patents, L. D. Gale, in 'On the relations of the American patent system to the progress of science', *Proceedings of the American Association for the Advancement of Science* (1854), 8, 292–301.

22 T. Antisell, *The Manufacture of Photogenic or Hydro-Carbon Oils, from Coal and Other Bituminous Substances, Capable of Supplying Burning Fluid*, New York, 1859.

extended defence of Young's patent. Given its close attention to legal records, Antisell's book might also have been a defence of the scientific standards of the Patent Office. In 1836, US patent law had been reformed such that a Commissioner of Patents and a number of assistants were appointed to study each application for novelty and originality before granting a patent. Since the Commissioner and his assistants were usually respected members of the scientific community, American patents were in effect sealed with scientific approval, and any dispute would have to confront the initial scientific evaluation.²³

Antisell thought American hostility towards Young was unfounded. 'An impression had taken hold of the American manufacturing public', he observed, 'that the patent of James Young has no force, as it was not a new invention at the date of the patent; and from the unfavorable effect of that patent upon the actual manufacture of coal oils in this country an ill-feeling has been produced against it'.²⁴ Antisell allowed that Young had acted unwisely in pressuring Americans, but that did not undermine Young's legal rights, of which there was 'no shadow of a doubt' in his mind. To prove his point, Antisell presented a 'History of the Art', which covered more than a century of discoveries, both scientific and technological.²⁵

The writing of 'histories' and the writing of patents themselves are singularly constructivist moments – the fabrication of 'truth', in this instance the origins of coal oil.²⁶ Accordingly, Antisell began with a review of experiments on the distillation of coal, conducted sporadically throughout the seventeenth and eighteenth centuries.²⁷ Not until the early nineteenth century, he wrote, were chemical products of coal systematically investigated. Important experiments were conducted in the 1830s by the German chemist and *Naturphilosoph* Karl Reichenbach (1788–1869).²⁸ Unlike his predecessors, Reichenbach heated samples of coal gradually, which resulted in the production of greater quantities of liquid (as opposed to gas, produced from heating coal rapidly). After further distilling this coal liquid, Reichenbach isolated a white crystalline substance similar in appearance to wax, which he called paraffine. The oil in which this substance was suspended Reichenbach named paraffine oil. He published these results, but, according to Antisell, Reichenbach did not continue the research, nor did others pursue it. Subsequently,

23 For more on the US Patent Office see R. C. Post, *Physics, Patents, and Politics: A Biography of Charles Grafton Page*, New York, 1976; and "'Liberalizers" versus "Scientific Men" in the antebellum Patent Office', *Technology and Culture* (1976), 17, 24–54. Britain had no such law for the examination of patent applications; instead it operated a mere registration system. The system was simplified, but not substantially altered, by the Reform Act of 1852. See Dutton, op. cit. (6), 57–68.

24 Antisell, op. cit. (22), 15.

25 For very similar histories, also written by defenders of Young's patent, see B. H. Paul, 'On destructive distillation, considered in reference to modern industrial arts', read before the Society of Arts, 27 May 1863, reprinted in *Chemical News* (1863), 7, 282–3, 295–7, and (1863), 8, 56–8, 78–9; and E. Frankland, 'On artificial illumination', *Chemical News* (1863), 7, 91–3.

26 For an insightful discussion of patents as history see Geof Bowker, 'What's in a patent?', in *Shaping Technology/Building Society: Studies in Sociotechnical Change* (ed. W. E. Bijker and J. Law), Cambridge, 1992, 53–74.

27 Stephen Hales, for instance, had distilled Newcastle coal and produced gas, liquid and tar; see S. Hales, *Vegetable Staticks*, London, 1727, ch. 6: 'Analysis of the air'.

28 Reichenbach used coal from Moravian mines. See W. V. Farrar, 'Reichenbach, Karl (or Carl)', *DSB*, xi, 359–60.

it was forgotten during the 1840s. To Young, then, is given the credit of applying earlier scientific research to produce coal oil. In Antisell's words,

The practical manufacture of oils from coal is due to James Young, [and] really two great results were first demonstrated practically by the operation of Young's process, namely – 1st. That coal was a material from which liquids could be manufactured economically... and 2d. That the liquids so formed were paraffin-containing compounds.²⁹

The relevant part of Antisell's justification of Young's priority rested on the terms 'practically' and 'economically', and this signalled a significant shift in the debate: essence and principle of the invention were defined as profitable result. Antisell credited Reichenbach with the scientific discoveries of paraffine and paraffine oils, but Young's was a practical and, therefore, different discovery. In a word, Young's patent was valid.

To respond to Antisell a group of American manufacturers consulted the Boston chemist Frank Storer. As one of the younger generation of American scientists, Storer had received his early education at Harvard, and then sought advanced training in Europe. He spent two years (1855–57) with such noted chemists as Robert Bunsen in Heidelberg, Theodor Richter in Freiberg and Emile Kopp in Paris. Upon his return to America Storer established a private chemical laboratory and became a consultant for the Boston Gas Light Company.³⁰ Relying on his manufacturing experience and knowledge of European research, Storer delivered a strong challenge to Antisell in a remarkably hostile review of Antisell's book published in the *American Journal of Science* in July 1860.³¹ It might be an over-generalization to assume that Storer's views were shared by the entire spectrum of the American scientific elite; however, many readers, including Antisell, thought this was the case. The editors of the *American Journal of Science* had in fact endorsed Storer with a short introduction admonishing others to make a 'careful perusal' of this review for it would 'amply repay' the effort.³²

Storer began with a ruthless criticism of Antisell's 'most lamentable lack of familiarity with the chemistry of the subject' and noted that 'the book is simply a jumble of badly selected abstracts, huddled together in a manner which must be anything but edifying to the student'.³³ He levelled his harshest words against Antisell's 'History'. Following the best-practised approach to nullifying a patent, Storer tried to establish prior discovery and public knowledge of the coal oil process by pointing to the work of Reichenbach.³⁴ Whereas Antisell had refused to allow scientific research to pre-empt practical discovery, Storer sought to overturn this interpretation by adducing a series of articles, 'some twenty or more', that Reichenbach had published on the distillation of coal in leading German scientific journals of the day. Storer noted:

29 Antisell, op. cit. (22), 14–15.

30 'Storer, Francis Humphreys', *Dictionary of American Biography*, xviii, 94–5.

31 F. H. Storer, 'Review of Dr. Antisell's work on photogenic oils, &c', *American Journal of Science* (1860), 30, 112–21, 254–64.

32 B. Silliman, Jr and J. D. Dana, 'Introduction to Frank H. Storer, "Review of Dr. Antisell's Work on Photogenic Oils, &c"', *American Journal of Science* (1860), 30, 112.

33 Storer, op. cit. (31), 112.

34 Antisell actually thought the review was 'double handed', meaning 'the joint production of a young chemist and a more mature patent solicitor'. T. Antisell, 'A scientific reviewer reviewed', *American Gas-Light Journal* (16 July 1860), 2, 25–6.

Looking at the question for a moment, solely in its scientific bearings, we cannot refrain from an expression of astonishment, that the details of Reichenbach's researches are so little known to the generality of chemists; while on the other hand, we are forced to confess, that it is indeed rare that scientific researches, conducted by a chemist in his laboratory, have so fully described a future art – have so accurately pointed out the methods to be followed and precautions to be observed by the practical manufacturer.³⁵

Storer faulted Antisell and his generation of chemists for their ignorance of German research. Perhaps if Antisell had understood the language, he too would have realized that Reichenbach's scientific discovery was, in essence and principle, identical to Young's practical discovery. More important, since Reichenbach published his discovery, he had fulfilled one of the purposes of a patent, namely he made his discovery public. Priority belonged to the scientist, said Storer, hence Young's patent was void.

For Storer and American manufacturers, therefore, publication was the key to priority and ownership. In this regard, they were supported by legal theory which argued that one reason for the existence of patents was to encourage individuals to make their discoveries public, thereby enriching the nation while at the same time giving due credit and protection to the discoverer.³⁶ Sheltering behind Storer's defence, many Americans consequently refused to surrender royalties to Young.³⁷

THE BRITISH CONTEXT

The successful defence of the rights of American manufacturers may have emboldened their British counterparts to press on in spite of threats of court action. Although it is not known whether Americans played any direct role, Young suspected that they had financed some of his British rivals. A business partner of Young's predicted, 'I think it likely the Yankees will purchase all our opponents and thus set them on their legs.'³⁸ Regardless of the specific financial arrangements, it is clear that the coal oil controversy did not divide the troops entirely along national lines. One of Young's British competitors was the Clydesdale Chemical Company, and in the first week of November 1860 Young was in a Scottish court suing them for patent infringement.³⁹

The scientific and technological debates at the heart of the Clydesdale trial were nearly identical to those staked out by Antisell and Storer, and American and British

³⁵ Storer, *op. cit.* (31), 117.

³⁶ Dutton, *op. cit.* (6), 17–33.

³⁷ It must also be noted that Abraham Gesner played a role in the successful defence, at least in print, of the American manufacturers. See A. Gesner, *A Practical Treatise on Coal, Petroleum, and Other Distilled Oils*, New York, 1861.

³⁸ E. W. Binney to E. Meldrum, 11 January 1859, quoted in Butt, 'Legends', *op. cit.* (8), 25.

³⁹ E. W. Binney and Company *versus* The Clydesdale Chemical Company, 1–7 November (Court of Session, Edinburgh, 1860). The case was watched very closely in the United States for it was widely believed that its outcome would set a precedent for the prosecution of similar American cases. See 'Young's coal oil patent case', *Scientific American* (5 January 1861), 4, 11; and 'Coal oil patent – important case', *Scientific American* (8 December 1860), 3, 378. The *American Gas-Light Journal* published two accounts of the trial: 'Coal-oil. Important patent decision', *American Gas-Light Journal* (1 December 1860), 2, 172; and 'The paraffine patent case' *ibid.* (1 January 1861), 2, 206.

commentators were keenly aware of the parallel lines of argument. Thus, for example, when the defence of the Clydesdale Company advanced Storer's argument of scientific precedence, the *American Gas-Light Journal* delighted in ridiculing it as follows:

The defendants set up that the process described in the patent was well known to the scientific world before the alleged invention by Young. To support this, all the old fossil apothecary books, chemical works, encyclopedias, and French and English patents, of late so fatiguingly inflicted upon the public by Frank Storer and others, were dragged into Court, where a number of chemical experts (?) stood ready to swear, that out of the old, dry bones they could construct Mr. Young's living invention.⁴⁰

In this instance Storer served as a sort of straw man of science whose expertise and professional disinterestedness appeared rather dubious in the eyes of the public.

The patent case lasted six days and, in the opinion of one observer, 'the whole of chemical science was ransacked by the defendants'.⁴¹ For both defence and plaintiff the arguments were carried forward by scientists, and this in itself is an important development and one which has not received much attention from historians of science and technology.⁴² The recruitment of expert witnesses was an increasingly common feature of patent disputes of the mid-nineteenth century. On the one hand, the involvement of scientists was a reflection of the increasing complexity of patents, especially of chemical procedures. On the other hand, it spoke of a new public role of scientists – the legal consultant. Taken together, it meant that scientists were becoming arbiters of patent disputes.

In the case against the Clydesdale Company, the 'eminent chemists' Frederick Penny, Lyon Playfair and Sir Robert Kane appeared as expert witnesses for Young. They allowed that even if Reichenbach had discovered paraffine oil, he had not produced it in 'economic quantities'. For the Clydesdale Company, equally 'eminent chemists' Dugald Campbell, Alfred Swaine Taylor and William Thomas Brande swore that the process, regardless of quantity, was well known to science and had been published in numerous works; hence, it was public knowledge to which Young could not have proprietary rights.⁴³

In presenting his summation before the jury, the Lord President of the Court of Session, Duncan McNeill, asked it to consider:

40 'Coal oil. Important patent decision', op. cit. (39).

41 'Young's coal oil patent case', op. cit. (39).

42 See, however, R. Smith and B. Wynne (eds.), *Expert Evidence: Interpreting Science in the Law*, London, 1989; C. Hamlin, 'Scientific method and expert witnesses: Victorian perspectives on a modern problem', *Social Studies of Science* (1986), 16, 485–573; J. Z. Fullmer, 'Technology, chemistry, and the law in early 19th-century England', *Technology and Culture* (1980), 21, 1–28; and W. H. Brock, 'The spectrum of science patronage', in *Patronage of Science in the Nineteenth Century* (ed. G. L'E. Turner), Leyden, 1976, 173–206. Yet historians of medicine seem much more conscious of the relations between medicine and the law, as in C. Crawford, 'Medicine and the law', in *Companion Encyclopedia of the History of Medicine* (ed. W. F. Bynum and R. Porter), 2 vols., London, 1993, ii, 1619–40; M. Clark and C. Crawford (eds.), *Legal Medicine in History*, Cambridge, 1993; R. Smith, *Trial by Medicine: Insanity and Responsibility in Victorian Trials*, Edinburgh, 1981; C. G. Jones, *Expert Witnesses: Science, Medicine, and the Practice of Law*, Oxford, 1994; and J. C. Mohr, *Doctors and the Law: Medical Jurisprudence in Nineteenth-Century America*, New York, 1993.

43 William Thomas Brande (1788–1866), FRS, was the successor to Humphry Davy as Professor of Chemistry at the Royal Institution in 1813 and editor of the *Dictionary of Science and Arts* from 1842 till his death. Alfred Swaine Taylor (1806–80) held the first professorship of medical jurisprudence at Guy's Hospital (1831–80) as well as a lectureship in chemistry (1832–80); he wrote *Medical Jurisprudence* (1843), the standard text of the time.

whether there was anything, either in chemistry or in any other science, which might not be placed under some general law. It did not follow, however, although the general law were known, that its application in any particular instance was also known. The patentee admitted that it was known that [paraffine] products could be obtained from coal; but it was not known that paraffine in merchantable quantity, could be obtained from it – and that was his discovery.⁴⁴

Lord McNeill's argument was broadly the same as that used by Antisell: there was a clear and legal distinction between scientific knowledge and its application. Young's was a practical discovery by virtue of the amount of coal oil produced. In this way, Lord McNeill could reject the idea that scientific publication amounted to prior invention.

On 7 November, the jury returned a verdict for Young, and the judge awarded him £7500 in costs and damages. Young's small band of American supporters were pleased, to say the least. 'The present trial has proved', *Scientific American* boasted, 'that the views which we first expressed, and with which Dr. Antisell coincided, have been confirmed'. It called the trial 'one of the most important patent cases ever tried'.⁴⁵ Hyperbole aside, other American observers thought that 'the verdict may be regarded as finally and triumphantly establishing the validity of the [Young's] patent both in Great Britain and in [America]'.⁴⁶

Despite such confidence, Young's rights in either Britain or America were not entirely secure; there were still larger competitors infringing his patent. The British coal oil controversy reached its climax, and eventual closure, in 1864 in a very expensive and elaborate trial in the Courts of Chancery, James Young *versus* Ebenezer Fernie.⁴⁷ Scientific expertise and the question of priority again dominated the proceedings. This time Young introduced August Wilhelm Hofmann (1818–92), the acknowledged doyen of coal tar chemistry, as his principal witness.⁴⁸ Hofmann was questioned for three days, during which he discussed everything from mineralogical classifications of coal to recent chemical research. On the subject of Reichenbach, Hofmann agreed that the German chemist had discovered paraffine and even allowed that Reichenbach might have had 'a commercial article in view'. This extraordinary admission seemed to undermine any justification of Young's patent on the grounds of economic quantity. But Hofmann qualified his statement in a crucial way: Reichenbach's work was essentially 'scientific', and 'his experiments on the distillation of coal were laboratory experiments'. More important, Reichenbach, like Hofmann himself, was first and foremost a scientist. In contrast, Young was a 'practical chemist', who, in Hofmann's opinion, had discovered a commercial process. For

44 Cited in 'The paraffine patent case', op. cit. (39).

45 'Young's coal oil patent case', op. cit. (39).

46 'Coal-oil. Important patent decision', op. cit. (39).

47 James Young and Others *versus* Ebenezer Fernie and Others, 29 February – 7 May (Courts of Chancery, 1864), reported in 'Young *versus* Fernie', *The Law Times Reports*, op. cit. (10). The trial received an extended review in the London *Chemical News*. See 'Young *v.* Fernie', *Chemical News* (21 May 1864), 9, 249–50; (28 May 1864), 9, 262–4; and (4 June 1864), 9, 273–6.

48 Hofmann was the first director of the Royal College of Chemistry (1845–63), and the most successful chemical expert of his generation; he annually augmented his income by £8000–£9000 as a legal consultant, see Brock, op. cit. (42), 186. At the same time as he was defending Young, for example, Hofmann was the chief chemical expert for the plaintiff in the trial Renard *versus* Levinstein (Courts of Chancery, 1864), a patent dispute over a blue aniline dye. See 'Renard *v.* Levinstein', *Chemical News* (9 April 1864), 9, 168. For more on Hofmann's career see Bud and Roberts, op. cit. (7); Beer, op. cit. (4); and Travis, 'Science's powerful companion', op. cit. (5).

Hofmann, the essential difference lay in the occupations of the two individuals (scientist and entrepreneur) and that decided the issue of types of knowledge.⁴⁹

In deciding the case, the Vice-Chancellor, Sir John Stuart, relied on Hofmann's testimony and then turned to the book by the 'eminent American chemist, Dr. Antisell, who holds an important position in the Patent Office of the United States of America'. There, Stuart found what he thought was the most elegant exposition of the principles upon which to decide the case. According to Stuart:

'So remained paraffine until this hour in the collection of chemical preparations, but it has never escaped from the rooms of the scientific man.'⁵⁰ Something therefore remained to be ascertained in order to [make] the useful application of this article for economical and commercial purposes. This illustrates the important distinction between the discoveries of the merely scientific chemist and of the practical manufacturer who invents the means of producing in abundance, suitable for economical and commercial purposes, that which previously existed as a beautiful item in the cabinets of men of science.⁵¹

The difference between scientific and technological knowledge, for him, was in the intention of the knowledge-producer. Merely scientific knowledge remained inert in cabinets of curiosity, while practical knowledge was to be used. Young's patent was valid by virtue of its commercial intention, and Stuart ordered Ebenezer Fernie to pay costs amounting to £10,000 and damages of £11,422. It was the largest sum of money Young was ever awarded, and it marked the end of the coal oil controversy in Britain.

INDUSTRIAL CONSEQUENCES

In the United States, the coal oil controversy was never truly settled as it was in Britain because Young's legal suits apparently never came to trial. Part of the reason for this was the published scientific arguments of Storer, but other factors external to the controversy must also be taken into account. The American Civil War, for one, completely disrupted the pre-existing coal oil industry, especially in the South and in the border states of Tennessee, Kentucky and West Virginia. Another factor was the petroleum boom in western Pennsylvania. Beginning in the summer of 1859, petroleum, or crude coal oil as it was initially called, quickly overtook coal as the raw material from which to refine illuminants. By 1862 over a million barrels were being produced at prices far below those of coal oil. Given its price and abundance, manufacturers of illuminating oils had a great incentive to switch to petroleum. Besides, no scientific or practical chemist then held a patent or could defend a claim to the process for refining petroleum. By the early 1860s, Kerosene, now a label for a petroleum-based illuminant, had replaced Kerosene, the coal oil. Illuminating oils distilled from coal disappeared entirely from the American market.

This fact was made particularly clear in 1864 when Young, flush from his victory over Ebenezer Fernie, applied for an extension of his American patent. He was granted it

49 'Young v. Fernie', *Chemical News*, op. cit. (47), 263.

50 Stuart was directly quoting Antisell, op. cit. (22), 14.

51 'Young v. Fernie', *Chemical News*, op. cit. (47), 276; and 'Young versus Fernie', *The Law Times Reports*, op. cit. (10), 865.

without opposition because there were no longer any coal oil manufacturers in the United States.⁵² In that same year, Young made a second visit to the United States, and he toured the petroleum regions of western Pennsylvania. According to one observer he was 'somewhat alarmed' at the thriving developments.⁵³ Ironically, then, Young's persistent threat of patent litigation might have helped to persuade American manufacturers to switch to petroleum, and in this way Young unwittingly encouraged industrial innovation. In contrast, Young's success in the British courts might explain why coal oil continued to be produced in Britain throughout the nineteenth and twentieth centuries, as recently as the early 1970s. It was only with Britain's own petroleum boom in the North Sea that coal oil manufacturing became redundant.

THE DIFFERENCE BETWEEN SCIENTISTS AND INVENTORS

The decisions of the British courts were made on economic grounds, but they were based in large part on assumptions about the traditional social roles of scientists and inventors. Since the seventeenth century scientists have assigned themselves the disinterested task of investigating and experimenting on nature. Likewise, inventors and practical men have assumed the job of pursuing production and profit. The distinction was so familiar by the mid-nineteenth century as to be unnecessary of explanation. Nevertheless, the chemist Edward Frankland, in a lecture before the Royal Institution in 1863 on the subject of artificial illumination, decided to make it plain for his listeners. He too agreed that Reichenbach had discovered paraffine, but Reichenbach was, in Frankland's words, a 'man of science'. Thus,

It is no part of his duty, it is not his function, to apply those [scientific] truths to the utilities of life. Success in this direction demands quite different powers of mind.⁵⁴

For Frankland not only were the social functions and practices of scientists and inventors different, but the intrinsic psychological make-up of members of each group was distinct.⁵⁵

In his address on the 'Encouragement of the Sciences', the Bostonian Charles Jackson had made a similar argument, but in a more colourful way:

No true man of science will ever disgrace himself by asking for a patent; and if he should, he might not know what to do with it any more than the man did who drew an elephant at a raffle. He cannot and will not leave his scientific pursuits to turn showman, mechanic, or merchant.⁵⁶

For Jackson it was not so much a question of temperament as it was professional ethics. Scientists must be above commercial interests. To confound the search for knowledge with the pursuit of profit would compromise the purity of science.

52 Williamson and Daum, op. cit. (13); and P. Lucier, 'Petroleum: what is it good for?', *American Heritage of Invention and Technology* (1991), 7, 56–63.

53 J. Merrill, 'Reminiscences', in *Derrick's Hand-Book of Petroleum: A Complete Chronological and Statistical Review of Petroleum Developments from 1859 to 1898*, Oil City, PA, 1898, 880–90, on 882.

54 Frankland, op. cit. (25), reprinted in *Journal of the Franklin Institute* (1864), 48, 30–7.

55 For a comparison of scientific and technological norms see Edwin T. Layton, Jr, 'Mirror-image twins: the communities of science and technology in 19th century America', *Technology and Culture* (1971), 12, 562–80, and D. J. DeSolla Price, 'Is technology historically independent of science?', *ibid.*, (1965), 6, 553–68.

56 Jackson, op. cit. (1), 237.

It also threatened the traditional domain of inventors, who were the most vociferous critics of Storer's attempt to rest priority in scientific publications. One writer using the pseudonym 'Paraffine' made a typical argument as follows:

It is a well known fact that the science of chemistry is rapidly becoming contemptible in the community, or at least among such portion of the people as are aware of the undeniable fact, that no new and useful invention has ever been made and patented that involved the discovery or new and useful application of anything appertaining to chemistry, that has not been attacked by professional chemists of apparently high standing among their brethren, and attempted to be overthrown on the score of want of novelty.⁵⁷

To equate scientific publication with invention meant interference and competition to artisans and practical men. Scientists should have nothing to do with invention and innovation.

Antisell was in complete accord with inventors. In a scathing article published in the *American Gas-Light Journal*, he denounced Storer's 'neoteric intolerance', a not uncommon fault among scientists in general, he said, but especially true of young, European-trained chemists who tended to evaluate past inventions by the scientific knowledge of the present day. Such arguments were as familiar as they were fallacious. To Antisell's mind, technological invention often preceded scientific discovery, and usually scientific laws were only *post hoc* explanations of already existing manufacturing practices. To grant scientific publication priority was to steal the discovery from the inventor.⁵⁸

Further, Antisell accused Storer and the editors of the *American Journal of Science* of compromising the ethical standards of science. He thought Storer's review was 'unworthy of a high-toned journal', and warned:

It is a matter of grave consideration...to find the earliest and the only professedly pure scientific journal departing from its legitimate functions and prostituting its columns, for an object which...does not come within the true sphere of its duties.⁵⁹

Storer had not acted the part of disinterested observer and, what was worse, neither had the *American Journal of Science*; both had 'prostituted' themselves to commercial interests.

The public face of science was likewise bloodied by journalistic attacks on those eminent authorities who had participated in Young's court cases. Because the disagreement among the expert witnesses had made it into the press, it raised serious professional concerns. The *Chemical News* observed:

Assuredly Science in the lecture-theatre and Science in the witness-box are two distinct beings. Of all sciences, perhaps Chemistry has, hitherto, cut the poorest figure in the courts of justice. The men most distinguished in their several departments are called upon to swear to what appears a simple matter of fact.⁶⁰

But witness-box quarrelling seemed to reduce scientific authority to mere opinion. The *Chemical News* summed up the coal oil controversy this way:

57 Paraffine, 'Coal-oil', *American Gas-Light Journal* (15 September 1860), 2, 86.

58 Antisell, op. cit. (34).

59 Antisell, op. cit. (34).

60 'The Torbanehill mineral', *Chemical News* (2 March 1861), 3, 217.

The evidence of 'Experts' is just now the object of general derision. Smart newspaper writers, wishing to indite a telling article, select the discrepancies in scientific evidence for a theme; noble Lords, anxious to enliven the dull debates of our hereditary legislators, find nothing so provocative of laughter as a story about the differences of 'mad doctors;' and barristers, ready to advocate any opinion, and anxious, perhaps, for a monopoly of the 'any-sidedness,' when addressing a jury, dilate with well-simulated indignation on the fact that eminent scientific men are to be in the witness-box on opposite sides.⁶¹

Science in the courtroom threatened to split the professional community.⁶² Patent disputes not only divided experts on simple matters of fact, they seemed to undermine their ethical standards. Involvement in industry caused considerable commotion within the ranks of British and American scientists.

In a way this was nothing unusual. Since the seventeenth century commercial interests had threatened scientists' claims to the disinterested pursuit of natural knowledge.⁶³ Edward Frankland and Charles Jackson had merely characterized ideal types, to which professionals should aspire. But clearly not all scientists subscribed to the normative vision. The *Scientific American* made this point in its review of Jackson's address. 'What kind of view has [Jackson] of what he calls a scientific man?', it asked its readers. 'Is it a disgrace to be mechanic and merchant, and exhibit a patent?' The answer was a resounding no, for 'quite a number of scientific men' had taken out patents. The *Scientific American* thought that Jackson had the 'wrong idea' of the modern scientist.⁶⁴

By the mid-nineteenth century there were plenty of examples of scientists who willingly patented their discoveries. Gesner, the inventor of Kerosene, was one, and his patents and publications supported Storer and the American manufacturers against Young. William Thomson, Lord Kelvin, was another and far more famous example.⁶⁵ Scientists obviously could and did patent their discoveries, and this fact surfaced during Young's patent cases. No one seriously doubted that Reichenbach could have patented paraffine and paraffine oils. But Reichenbach chose not to, thereby reaffirming his standards as a professional scientist and opening the door to Young.

CONCLUSION

The coal oil controversy clearly shows that the relations of science and industry in the mid-nineteenth century were contentious and problematic. Scientists and inventors each had their self-designated and socially attributed roles. The boundaries between the two groups

61 'The evidence of experts', *Chemical News* (5 April 1862), 5, 183.

62 For further discussion of the effects of expert witnessing on the practice, public image and ethics of scientists see Hamlin, op. cit. (42).

63 Disinterestedness, of course, was one of the basic institutional elements of science for Robert K. Merton; see R. K. Merton, 'The normative structure of science', *The Sociology of Science*, Chicago, 1973, 267–78. Larry Stewart has made a persuasive case for the close interrelations of science and commerce in the seventeenth and eighteenth centuries; see L. Stewart, *The Rise of Public Science: Rhetoric, Technology, and Natural Philosophy in Newtonian Britain, 1660–1750*, Cambridge, 1992. Also see P. Dear, 'From truth to disinterestedness in the seventeenth century', *Social Studies of Science* (1992), 22, 619–31; G. Tweedale, 'Geology and industrial consultancy: Sir William Boyd Dawkins (1837–1929) and the Kent Coalfield', *BJHS* (1991), 24, 435–51; and Lucier, op. cit. (5), 264–7.

64 'Dr. Jackson's Address before the American Institute', *Scientific American* (1 November 1851), 7, 51.

65 For more on Kelvin's commercial interests see C. Smith and M. N. Wise, *Energy and Empire: A Biographical Study of Lord Kelvin*, Cambridge, 1989.

were well guarded. Any scientist who patented research put at risk his professional integrity. Still, if a scientist wanted to protect his rights as a discoverer, he would have to patent. Publication was not sufficient on its own to ensure proprietary rights.

In this regard, the coal oil controversy revealed that the distinction in types of knowledge – science versus technology, research versus applied science – was fundamentally arbitrary. Legal differences between scientific discovery and technological principle could be maintained only by recourse to a fiction of commercial intent of the knowledge-producer. In essence, the debate had turned. If there were no differences in types of knowledge, then all that remained were the social demarcations created and maintained by the knowledge-producers themselves. The integration of science and industry in the late nineteenth century thus is a story about changes in social relations, not about the sudden usefulness of science itself.